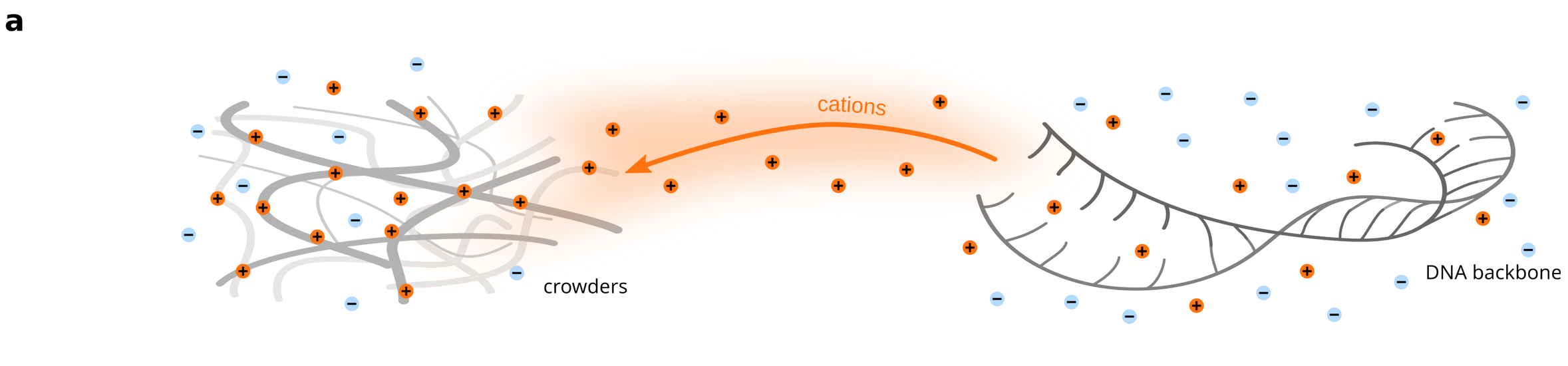

a
cations
crowders
DNA backbone
b
DOSY signal decay for PEG 400
PEG 400, ^{1}H, D = (0.42 ± 0.01)·10^{-9} m^2/s
Na$^+$ in PEG 400, ^{23}Na, D = (1.48 ± 0.08)·10^{-9} m^2/s
normalized intensity I/I_0
gradient strength [G/cm]
c
$D_{crowder}$ vs crowder wt.%
EG, R^2 = 0.979
PEG 200, R^2 = 0.995
PEG 400, R^2 = 0.998
PEG 600, R^2 = 0.996
PEG 1k, R^2 = 0.981
PEG 1.5k, R^2 = 0.982
PEG 3k, R^2 = 0.999
PEG 6k, R^2 = 1.000
PEG 12k, R^2 = 0.999
PEG 20k, R^2 = 1.000
PEG 35k, R^2 = 0.999
$D_{crowder}$ [um^2/s]
crowder wt.%
d
D_{Na^+} vs crowder wt.%
EG
PEG 200
PEG 400
PEG 600
PEG 1k
PEG 1.5k
PEG 3k
PEG 6k
PEG 12k
PEG 20k
PEG 35k
buffer
D_{Na^+} [um^2/s]
crowder wt.%
e
nanoviscosity effect for PEG 35k
nanoviscosity effect
D_{Na^+} measured
D_{Na^+} [um^2/s]
crowder wt.%
f
βn fit for EG and PEG 35k
EG fit: $\beta n = 0.43 \pm 0.07$ M^{-1}
PEG 35k fit: $\beta n = 143.24 \pm 8.69$ M^{-1}
EG data
PEG 35k data
buffer
D_w [μm^2/s]
crowder wt.%
g
βn vs degree of polymerization
fit: $\beta n = 0.175$ DP
EG
PEG 200
PEG 400
PEG 600
PEG 1k
PEG 1.5k
PEG 3k
PEG 6k
PEG 12k
PEG 20k
PEG 35k
βn [M^{-1}]
degree of polymerization